

CLASSIFICATION OF $\mathbb{E}_n$ -MAPS FROM $\mathbb{Z}$ TO $\mathrm{Pic}(\mathbb{S})$

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ABSTRACT. Regard the ordinary abelian group $(\mathbb{Z}, +)$ as a discrete grouplike $\mathbb{E}_\infty$ -space, and let $\mathrm{Pic}(\mathbb{S})$ be the Picard space of the sphere spectrum. For every finite $n \geq 1$, we determine the full derived mapping space of unital $\mathbb{E}_n$ -maps $\mathbb{Z} \rightarrow \mathrm{Pic}(\mathbb{S})$. The answer is a canonical homotopy fiber obtained from the truncation fiber sequence of the Picard spectrum. The possible degrees are all integers for $n = 1$, the even integers for $n = 2$, and only zero for $n \geq 3$. The existence of a degree-two $\mathbb{E}_2$ -map is due to Lurie. Nonzero degree for $n \geq 3$ is excluded by a Thom-spectrum argument and the odd-primary Dyer–Lashof obstruction of Rognes–Sagave–Schlichtkrull. We also treat $\mathbb{E}_\infty$ -maps. The classification is at the level of mapping spaces; the residual degree-zero information is expressed as $\mathrm{gl}_1(\mathbb{S})$ -cohomology of Eilenberg–Mac Lane spaces.

1. INTRODUCTION AND STATEMENT OF RESULTS

Lurie constructed, in the course of proving rotation invariance in algebraic K -theory, an $\mathbb{E}_2$ -monoidal functor

$$\Phi: \mathbb{Z} \longrightarrow \mathrm{Pic}(\mathbb{S})$$

whose value on an integer m is the even suspension $\mathbb{S}^{2m}$; see [4, Remark 3.5.10]. This raises the natural question of describing all $\mathbb{E}_n$ -maps from $\mathbb{Z}$ to the Picard space of the sphere, for arbitrary n . There are two logically distinct parts to the problem. One must determine the suspension degree of the image of the generator, and one must also retain the higher coherent choices that remain after that degree is fixed. The purpose of this paper is to give a uniform answer to both parts.

We write $\mathcal{S}$ for the ∞ -category of spaces, Sp for the ∞ -category of spectra, and $\mathbb{S}$ for the sphere spectrum. All $\mathbb{E}_n$ -algebras and morphisms are unital. Throughout, $\mathbb{Z}$ denotes the ordinary additive group of integers, regarded as a commutative group object in discrete spaces and hence as a grouplike $\mathbb{E}_\infty$ -space. Equivalently,

$$\mathbb{Z} \simeq \Omega^\infty H\mathbb{Z}$$

as grouplike $\mathbb{E}_\infty$ -spaces. We put $\mathbb{N} = \{0, 1, 2, \dots\}$, regarded as the ordinary discrete commutative monoid under addition. The Picard space $\mathrm{Pic}(\mathbb{S})$ is

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the maximal subgroupoid of $\mathbf{Sp}$ spanned by the invertible spectra, with multiplication induced by smash product.

Every invertible spectrum is a suspension of the sphere, and hence

$$\pi_0 \operatorname{Pic}(\mathbb{S}) \cong \mathbb{Z}, \quad [\mathbb{S}^d] \mapsto d;$$

see [5, Example 2.1.6]. If $f: \mathbb{Z} \rightarrow \operatorname{Pic}(\mathbb{S})$ is an $\mathbb{E}_n$ -map, the induced homomorphism on components is multiplication by a unique integer. We call this integer the *degree* of f :

$$\deg(f) = d \iff f(1) \simeq \mathbb{S}^d.$$

Let $\operatorname{pic}(\mathbb{S})$ denote the connective Picard spectrum and $\operatorname{gl}_1(\mathbb{S})$ the connective spectrum of units. The Postnikov truncation of $\operatorname{pic}(\mathbb{S})$ gives a canonical fiber sequence

$$\Sigma \operatorname{gl}_1(\mathbb{S}) \longrightarrow \operatorname{pic}(\mathbb{S}) \xrightarrow{\deg} H\mathbb{Z} \xrightarrow{\kappa} \Sigma^2 \operatorname{gl}_1(\mathbb{S}). \quad (1.1)$$

For $r \geq 0$, set

$$B^r \operatorname{Pic}(\mathbb{S}) := \Omega^\infty \Sigma^r \operatorname{pic}(\mathbb{S}), \quad B^r \operatorname{GL}_1(\mathbb{S}) := \Omega^\infty \Sigma^r \operatorname{gl}_1(\mathbb{S}),$$

and write

$$\kappa_n := \Omega^\infty \Sigma^n \kappa: K(\mathbb{Z}, n) \longrightarrow B^{n+2} \operatorname{GL}_1(\mathbb{S}).$$

Here $\Sigma^\infty X$ denotes the reduced suspension spectrum of a based space X , and $F(-, -)$ denotes the internal mapping spectrum. For a spectrum E , a based space X , and an integer q , we use the cohomological grading convention

$$\tilde{E}^q(X) := [\Sigma^\infty X, \Sigma^q E]_{\mathbf{Sp}}. \quad (1.2)$$

We write $\mathbb{Z}_{\text{disc}}$ for the discrete space whose underlying set is $\mathbb{Z}$.

For finite $n \geq 1$, define

$$\mathcal{M}_n := \operatorname{Map}_{\operatorname{Alg}_{\mathbb{E}_n}}(\mathbb{S}, \operatorname{Pic}(\mathbb{S})), \quad \mathcal{Y}_n := \operatorname{Map}_*(K(\mathbb{Z}, n), B^{n+2} \operatorname{GL}_1(\mathbb{S})),$$

and

$$\mathcal{T}_n := \operatorname{Map}_*(K(\mathbb{Z}, n), B^{n+1} \operatorname{GL}_1(\mathbb{S})). \quad (1.3)$$

Theorem 1.1 (Classification for finite n). *For every finite $n \geq 1$, the following statements hold.*

(i) *There are natural equivalences*

$$\mathcal{M}_n \simeq \operatorname{Map}_*(K(\mathbb{Z}, n), B^n \operatorname{Pic}(\mathbb{S})), \quad (1.4)$$

$$\simeq \Omega^\infty F(\Sigma^\infty K(\mathbb{Z}, n), \Sigma^n \operatorname{pic}(\mathbb{S})). \quad (1.5)$$

(ii) *There is a canonical equivalence*

$$\mathcal{M}_n \simeq \operatorname{hofib}_0\left(\mathbb{Z}_{\text{disc}} \xrightarrow{d \mapsto d\kappa_n} \mathcal{Y}_n\right). \quad (1.6)$$

Consequently, an $\mathbb{E}_n$ -map of degree d exists if and only if $d\kappa_n$ is null-homotopic.

- (iii) Let $\mathcal{M}_n^{(d)}$ be the homotopy fiber of the degree map over $d \in \mathbb{Z}$. There is a canonical description

$$\mathcal{M}_n^{(d)} \simeq \text{Path}_{\mathcal{Y}_n}(d\kappa_n, 0), \quad (1.7)$$

where the right-hand side is the space of paths from $d\kappa_n$ to the basepoint. Thus every nonempty degree fiber is a torsor for

$$\Omega\mathcal{Y}_n \simeq \mathcal{T}_n \simeq \Omega^\infty F(\Sigma^\infty K(\mathbb{Z}, n), \Sigma^{n+1} \text{gl}_1(\mathbb{S})).$$

After choosing one point of the fiber, the torsor becomes equivalent to $\mathcal{T}_n$.

- (iv) The image of the degree map on components is

$$D_n := \text{im}(\pi_0 \mathcal{M}_n \xrightarrow{\deg} \mathbb{Z}) = \begin{cases} \mathbb{Z}, & n = 1, \\ 2\mathbb{Z}, & n = 2, \\ \{0\}, & n \geq 3. \end{cases} \quad (1.8)$$

- (v) More explicitly,

$$\mathcal{M}_1 \simeq \coprod_{d \in \mathbb{Z}} B \text{GL}_1(\mathbb{S}). \quad (1.9)$$

After choosing Lurie's degree-two map Φ , there is an equivalence

$$\mathcal{M}_2 \simeq \coprod_{k \in \mathbb{Z}} \mathcal{T}_2, \quad (1.10)$$

whose k -th summand is the degree- $2k$ fiber. For every $n \geq 3$, the constant map gives a canonical equivalence

$$\mathcal{M}_n \simeq \mathcal{T}_n. \quad (1.11)$$

The word “classification” in theorem 1.1 refers to the complete derived mapping space, not only to its set of components. It does not mean that the remaining generalized cohomology groups have been reduced to numerical tables. In particular, for $n \geq 3$, degree zero need not imply nullhomotopy.

Corollary 1.2 (Homotopy classes and higher homotopy groups). *Put*

$$G_n := \pi_0 \mathcal{T}_n = \widetilde{\text{gl}_1(\mathbb{S})}^{n+1}(K(\mathbb{Z}, n)).$$

Via the infinite-loop-space structure supplied by (1.5), $\pi_0 \mathcal{M}_n$ is an abelian group. After choosing Lurie's degree-two map, there are group isomorphisms

$$\pi_0 \mathcal{M}_n \cong \begin{cases} \mathbb{Z}, & n = 1, \\ 2\mathbb{Z} \oplus G_2, & n = 2, \\ G_n, & n \geq 3. \end{cases} \quad (1.12)$$

Moreover, for every $j \geq 0$,

$$\pi_j \mathcal{T}_n \cong \widetilde{\text{gl}_1(\mathbb{S})}^{n+1-j}(K(\mathbb{Z}, n)). \quad (1.13)$$

There is an analogous statement for infinite loop maps.

Theorem 1.3 (The $\mathbb{E}_\infty$ -case). *Let*

$$\mathcal{M}_\infty := \mathrm{Map}_{\mathrm{Alg}_{\mathbb{E}_\infty}(\mathcal{S})}(\mathbb{Z}, \mathrm{Pic}(\mathbb{S})).$$

Then every $\mathbb{E}_\infty$ -map has degree zero, and there are natural equivalences

$$\mathcal{M}_\infty \simeq \Omega^\infty F(H\mathbb{Z}, \mathrm{pic}(\mathbb{S})), \quad (1.14)$$

$$\simeq \Omega^\infty F(H\mathbb{Z}, \Sigma \mathrm{gl}_1(\mathbb{S})). \quad (1.15)$$

The proof has three ingredients. Recognition identifies $\mathbb{E}_n$ -maps between grouplike $\mathbb{E}_n$ -spaces with based maps between their n -fold deloopings. The truncation sequence (1.1) then gives the universal obstruction $d\kappa_n$. The remaining work is to determine which degrees occur. The cases $n = 1$ and $n = 2$ follow from freeness, the Koszul sign, and Lurie's construction. For $n \geq 3$, a nonzero-degree map would produce an $\mathbb{E}_3$ -structure on a free associative sphere algebra; the odd-primary obstruction of Rognes–Sagave–Schlichtkrull rules this out.

2. THE PICARD SPECTRUM AND THE UNIVERSAL OBSTRUCTION

2.1. The degree extension of the Picard spectrum.

Lemma 2.1 (The Picard spectrum of the sphere). *The Picard space $\mathrm{Pic}(\mathbb{S})$ is a grouplike $\mathbb{E}_\infty$ -space and is the zeroth space of a connective spectrum $\mathrm{pic}(\mathbb{S})$. There are canonical identifications*

$$\pi_0 \mathrm{pic}(\mathbb{S}) \cong \mathbb{Z}, \quad \tau_{\geq 1} \mathrm{pic}(\mathbb{S}) \simeq \Sigma \mathrm{gl}_1(\mathbb{S}).$$

Consequently, (1.1) is the canonical Postnikov fiber sequence of $\mathrm{pic}(\mathbb{S})$.

Proof. For a symmetric monoidal ∞ -category $\mathcal{C}$, its invertible objects and equivalences form a grouplike $\mathbb{E}_\infty$ -space. The equivalence between grouplike commutative monoids in spaces and connective spectra therefore defines $\mathrm{pic}(\mathcal{C})$; see [5, Definition 2.2.1] and [3, Section 1.4, especially p. 132]. For $\mathcal{C} = \mathrm{Sp}$, every invertible spectrum is a suspension of $\mathbb{S}$, so $\pi_0 \mathrm{pic}(\mathbb{S}) \cong \mathbb{Z}$; see [5, Example 2.1.6].

The loop space of the unit component of $\mathrm{Pic}(\mathbb{S})$ is $\mathrm{Aut}_{\mathrm{Sp}}(\mathbb{S}) = \mathrm{GL}_1(\mathbb{S})$. Equivalently, $\tau_{\geq 1} \mathrm{pic}(\mathbb{S}) \simeq \Sigma \mathrm{gl}_1(\mathbb{S})$; see [5, p. 3 and Example 2.2.2]. Since $\tau_{\leq 0} \mathrm{pic}(\mathbb{S}) \simeq H\mathbb{Z}$, the standard truncation sequence

$$\tau_{\geq 1} \mathrm{pic}(\mathbb{S}) \longrightarrow \mathrm{pic}(\mathbb{S}) \longrightarrow \tau_{\leq 0} \mathrm{pic}(\mathbb{S}) \longrightarrow \Sigma \tau_{\geq 1} \mathrm{pic}(\mathbb{S})$$

is precisely (1.1). □

Remark 2.2. The same references give

$$\pi_1 \mathrm{pic}(\mathbb{S}) \cong \{\pm 1\} \cong \mathbb{Z}/2, \quad \pi_i \mathrm{pic}(\mathbb{S}) \cong \pi_{i-1} \mathbb{S} \quad (i \geq 2).$$

These homotopy groups alone do not determine the mapping spaces in theorem 1.1; the extension class κ is essential.

Lemma 2.3 (The delooped degree sequence). *For every $n \geq 1$, applying $\Omega^\infty \Sigma^n$ to (1.1) gives a fiber sequence of infinite loop spaces*

$$B^{n+1} \mathrm{GL}_1(\mathbb{S}) \longrightarrow B^n \mathrm{Pic}(\mathbb{S}) \xrightarrow{\deg} K(\mathbb{Z}, n) \xrightarrow{\kappa_n} B^{n+2} \mathrm{GL}_1(\mathbb{S}). \quad (2.1)$$

Proof. Suspension is exact in spectra, Ω^∞ preserves limits, and $\Omega^\infty \Sigma^n H\mathbb{Z} \simeq K(\mathbb{Z}, n)$. Thus (2.1) is the image of (1.1) under $\Omega^\infty \Sigma^n$. $\square$

2.2. Recognition and the homotopy-fiber formula.

Proposition 2.4 (Recognition of the mapping space). *For every finite $n \geq 1$, there are natural equivalences*

$$\mathcal{M}_n \simeq \mathrm{Map}_*(K(\mathbb{Z}, n), B^n \mathrm{Pic}(\mathbb{S})) \simeq \Omega^\infty F(\Sigma^\infty K(\mathbb{Z}, n), \Sigma^n \mathrm{pic}(\mathbb{S})).$$

Proof. The recognition theorem, in the ∞ -categorical form of [3, Theorem 5.2.6.10], identifies grouplike $\mathbb{E}_n$ -spaces with pointed n -connective spaces; the inverse is the n -fold bar construction. Both $\mathbb{Z}$ and $\mathrm{Pic}(\mathbb{S})$ are grouplike, and their chosen n -fold deloopings are

$$B^n \mathbb{Z} \simeq \Omega^\infty \Sigma^n H\mathbb{Z} = K(\mathbb{Z}, n), \quad B^n \mathrm{Pic}(\mathbb{S}) = \Omega^\infty \Sigma^n \mathrm{pic}(\mathbb{S}).$$

An equivalence of ∞ -categories preserves derived mapping spaces, so this proves the first equivalence. The second is the based adjunction

$$\mathrm{Map}_*(X, \Omega^\infty E) \simeq \Omega^\infty F(\Sigma^\infty X, E)$$

for a based space X and a spectrum E . $\square$

Lemma 2.5 (Self-maps of $K(\mathbb{Z}, n)$). *For every $n \geq 1$,*

$$\mathrm{Map}_*(K(\mathbb{Z}, n), K(\mathbb{Z}, n)) \simeq \mathbb{Z}_{\mathrm{disc}}.$$

Under this equivalence, $d \in \mathbb{Z}$ corresponds to multiplication by d .

Proof. Let $X = K(\mathbb{Z}, n)$. Since the target is an abelian infinite loop space, translation identifies all components of the based mapping space. At the constant map, for $i \geq 0$,

$$\pi_i \mathrm{Map}_*(X, K(\mathbb{Z}, n)) \cong [\Sigma^i X, K(\mathbb{Z}, n)]_* \cong \tilde{H}^{n-i}(X; \mathbb{Z}).$$

For $i = 0$, this group is $H^n(K(\mathbb{Z}, n); \mathbb{Z}) \cong \mathbb{Z}$, generated by the universal class. It vanishes for $0 < i < n$ because X is $(n-1)$ -connected, for $i = n$ because $\tilde{H}^0(X; \mathbb{Z}) = 0$, and for $i > n$ by degree considerations. Hence the mapping space is discrete with components indexed by $\mathbb{Z}$. The component indexed by d multiplies the universal class by d , and is therefore the multiplication-by- d map. $\square$

Proposition 2.6 (The canonical obstruction and its fibers). *For every finite $n \geq 1$, there is a canonical equivalence*

$$\mathcal{M}_n \simeq \mathrm{hofib}_0\left(\mathbb{Z}_{\mathrm{disc}} \xrightarrow{d \mapsto d\kappa_n} \mathcal{Y}_n\right).$$

For each $d \in \mathbb{Z}$, the degree- d fiber is canonically the path space

$$\mathcal{M}_n^{(d)} \simeq \mathrm{Path}_{\mathcal{Y}_n}(d\kappa_n, 0).$$

It is nonempty precisely when $d\kappa_n$ is nullhomotopic. If nonempty, it is a principal homogeneous space, in the homotopy-coherent sense, under $\Omega\mathcal{Y}_n \simeq \mathcal{T}_n$.

Proof. Apply $\mathrm{Map}_*(K(\mathbb{Z}, n), -)$ to (2.1). Mapping out of a fixed space preserves limits, so $\mathrm{Map}_*(K(\mathbb{Z}, n), B^n \mathrm{Pic}(\mathbb{S}))$ is the homotopy fiber over the constant map of

$$\mathrm{Map}_*(K(\mathbb{Z}, n), K(\mathbb{Z}, n)) \longrightarrow \mathrm{Map}_*(K(\mathbb{Z}, n), B^{n+2} \mathrm{GL}_1(\mathbb{S})).$$

By lemma 2.5, the source is $\mathbb{Z}_{\mathrm{disc}}$. The point d is represented by multiplication by d on $K(\mathbb{Z}, n)$. Since $\kappa: H\mathbb{Z} \rightarrow \Sigma^2 \mathrm{gl}_1(\mathbb{S})$ is a map of spectra, postcomposition sends this point to the additive multiple $d\kappa_n$. The homotopy-fiber formula and the path-space description follow.

For any based space Y , the space $\mathrm{Path}_Y(y, 0)$ carries an action of ΩY by concatenation at the endpoint. If it is nonempty, choosing a path from y to 0 identifies it with ΩY , and any two such identifications differ by translation. Thus it is canonically an ΩY -torsor. Finally,

$$\Omega\mathcal{Y}_n \simeq \mathrm{Map}_*(K(\mathbb{Z}, n), B^{n+1} \mathrm{GL}_1(\mathbb{S})) = \mathcal{T}_n,$$

as claimed. $\square$

Remark 2.7 (The group structure on components). The function-spectrum model in proposition 2.4 equips $\mathcal{M}_n$ with an infinite-loop-space structure. Under this structure, the degree map is induced by a map of spectra and hence gives a homomorphism $\pi_0 \mathcal{M}_n \rightarrow \mathbb{Z}$. All group statements below refer to this canonical structure.

3. THE CASES $n = 1$ AND $n = 2$

3.1. The associative case.

Proposition 3.1. *There is a natural equivalence*

$$\mathcal{M}_1 \simeq \mathrm{Pic}(\mathbb{S}) \simeq \coprod_{d \in \mathbb{Z}} B \mathrm{GL}_1(\mathbb{S}).$$

In particular, every integer occurs as a degree.

Proof. By proposition 2.4,

$$\mathcal{M}_1 \simeq \mathrm{Map}_*(S^1, B \mathrm{Pic}(\mathbb{S})) \simeq \Omega B \mathrm{Pic}(\mathbb{S}) \simeq \mathrm{Pic}(\mathbb{S}).$$

Equivalently, $\mathbb{Z}$ is the free grouplike $\mathbb{E}_1$ -space on one generator, so that generator may be sent to any invertible spectrum. The components of $\mathrm{Pic}(\mathbb{S})$ are indexed by $d \in \mathbb{Z}$, and the automorphism space of $\mathbb{S}^d$ is $\mathrm{GL}_1(\mathbb{S})$. Therefore each component is $B \mathrm{GL}_1(\mathbb{S})$. $\square$

3.2. The braided case.

Lemma 3.2 (The symmetry sign for suspended spheres). *For integers a, b , the symmetry map*

$$\mathbb{S}^a \wedge \mathbb{S}^b \longrightarrow \mathbb{S}^b \wedge \mathbb{S}^a$$

represents $(-1)^{ab} \in \pi_0 \mathrm{GL}_1(\mathbb{S}) = \{\pm 1\}$, after the standard identifications of source and target with $\mathbb{S}^{a+b}$. In particular, the self-braiding of $\mathbb{S}^d$ is multiplication by $(-1)^d$.

Proof. The switch map $S^1 \wedge S^1 \rightarrow S^1 \wedge S^1 \cong S^2$ has degree -1 . For $a, b \geq 0$, identify S^a and S^b with smash powers of S^1 . Interchanging the two blocks requires ab adjacent switches and therefore has degree $(-1)^{ab}$. The braiding on the Picard groupoid is multiplicative in each variable. Since $\mathbb{S}^{-1}$ is the tensor inverse of $\mathbb{S}$, the resulting bicharacter extends uniquely from $\mathbb{N} \times \mathbb{N}$ to $\mathbb{Z} \times \mathbb{Z}$. Taking $a = b = d$ gives $(-1)^{d^2} = (-1)^d$. $\square$

Proposition 3.3 (Evenness of the degree). *Every $\mathbb{E}_2$ -map $f: \mathbb{Z} \rightarrow \mathrm{Pic}(\mathbb{S})$ has even degree.*

Proof. A path in the arity-two space of the little-two-cubes operad that exchanges the two cubes induces the braiding in the homotopy category of every $\mathbb{E}_2$ -algebra. A morphism of $\mathbb{E}_2$ -algebras preserves this braiding. In the discrete symmetric monoidal group $\mathbb{Z}$, the braiding of 1 with itself is the identity automorphism of 2. If $f(1) \simeq \mathbb{S}^d$, the self-braiding of $\mathbb{S}^d$ must consequently lie in the identity component of $\mathrm{Aut}(\mathbb{S}^{2d})$. By lemma 3.2, the class of this self-braiding in

$$\pi_0 \mathrm{Aut}(\mathbb{S}^{2d}) \cong \pi_0 \mathrm{GL}_1(\mathbb{S}) \cong \{\pm 1\}$$

is $(-1)^d$. Since the identity component represents 1, it follows that $(-1)^d = 1$, and hence d is even. $\square$

Proposition 3.4 (Lurie's degree-two map). *There exists an $\mathbb{E}_2$ -map*

$$\Phi: \mathbb{Z} \longrightarrow \mathrm{Pic}(\mathbb{S})$$

of degree 2. Consequently, every even integer occurs as the degree of an $\mathbb{E}_2$ -map.

Proof. Lurie constructs an $\mathbb{E}_2$ -monoidal functor from the ordinary discrete symmetric monoidal group $(\mathbb{Z}, +)$ to spectra that factors through $\mathrm{Pic}(\mathbb{S})$, and shows that its object function is

$$m \longmapsto \mathbb{S}^{2m};$$

see [4, Remark 3.5.10]. He identifies this functor with the composite of Bott periodicity and the complex J -homomorphism in [4, Corollary 7.2.5]. Thus $\deg(\Phi) = 2$.

For $k \in \mathbb{Z}$, multiplication by k is an $\mathbb{E}_\infty$ -endomorphism $[k]: \mathbb{Z} \rightarrow \mathbb{Z}$. The composite $\Phi \circ [k]$ is an $\mathbb{E}_2$ -map of degree $2k$. Together with proposition 3.3, this proves that the degree image is $2\mathbb{Z}$. $\square$

Corollary 3.5. *After choosing Φ , there is an equivalence*

$$\mathcal{M}_2 \simeq \coprod_{k \in \mathbb{Z}} \mathcal{T}_2,$$

where the summand indexed by k is the degree- $2k$ fiber.

Proof. By proposition 2.6, every nonempty degree fiber is a $\mathcal{T}_2$ -torsor. The maps $\Phi \circ [k]$ choose a point in every even fiber, and proposition 3.3 shows that no odd fiber is nonempty. Translation by these chosen points gives the stated equivalence. $\square$

4. THOM SPECTRA AND THE $\mathbb{E}_3$ -OBSTRUCTION

A nonzero-degree $\mathbb{E}_3$ -map would produce an $\mathbb{E}_3$ -ring structure on a free associative sphere algebra. We first identify the underlying associative Thom algebra without making any noncanonical coherent choices.

4.1. The associative Thom algebra over $\mathbb{N}$.

Lemma 4.1 (The free one-generator $\mathbb{E}_1$ -space). *The ordinary discrete monoid $(\mathbb{N}, +)$ is the free unital $\mathbb{E}_1$ -space on one point. Consequently, for every $\mathbb{E}_1$ -space A , evaluation at 1 induces an equivalence*

$$\mathrm{ev}_1: \mathrm{Map}_{\mathrm{Alg}_{\mathbb{E}_1}(\mathcal{S})}(\mathbb{N}, A) \xrightarrow{\simeq} A.$$

Proof. For a symmetric model of the little-one-cubes operad, the free $\mathbb{E}_1$ -space on a point is

$$\coprod_{j \geq 0} \mathbb{E}_1(j)_{h\Sigma_j}.$$

The space $\mathbb{E}_1(j)$ has $j!$ contractible components, freely and transitively permuted by Σ_j , so every homotopy-orbit space in the display is contractible. Operadic composition adds the integer j . Hence the free algebra is equivalent, as an $\mathbb{E}_1$ -space, to the ordinary discrete monoid $(\mathbb{N}, +)$. The evaluation equivalence is its universal mapping property. $\square$

Lemma 4.2 (The underlying associative Thom algebra). *Let $g: \mathbb{N} \rightarrow \mathrm{Pic}(\mathcal{S})$ be an $\mathbb{E}_1$ -map, and choose an equivalence $g(1) \simeq L$. Then its Thom spectrum*

$$Mg := \mathrm{colim}(\mathbb{N} \xrightarrow{g} \mathrm{Pic}(\mathcal{S}) \hookrightarrow \mathrm{Sp})$$

is equivalent, as an $\mathbb{E}_1$ -ring, to the free unital $\mathbb{E}_1$ -ring on L :

$$Mg \simeq \mathbb{E}_1(L) \simeq \bigvee_{j \geq 0} L^{\wedge j}. \quad (4.1)$$

If g is an $\mathbb{E}_k$ -map for $k \geq 1$, the Thom spectrum carries the canonical $\mathbb{E}_k$ -ring structure of [1, Corollary 3.2], and its restriction to $\mathbb{E}_1$ is the ring in (4.1).

Proof. Define the standard tensor-power functor

$$g_L: \mathbb{N} \longrightarrow \mathrm{Pic}(\mathcal{S}), \quad j \longmapsto L^{\wedge j},$$

where $L^{\wedge 0} = \mathbb{S}$, and equip it with the canonical coherently associative equivalences

$$L^{\wedge i} \wedge L^{\wedge j} \simeq L^{\wedge(i+j)}.$$

Thus g_L is an $\mathbb{E}_1$ -map whose value at 1 is L . By the evaluation equivalence of lemma 4.1, the chosen equivalence $g(1) \simeq L$ determines an equivalence

$$g \simeq g_L$$

in $\mathrm{Map}_{\mathrm{Alg}_{\mathbb{E}_1}(\mathcal{S})}(\mathbb{N}, \mathrm{Pic}(\mathbb{S}))$. The multiplicative Thom construction is functorial in $\mathbb{E}_1$ -maps to $\mathrm{Pic}(\mathbb{S})$; this is the operadic colimit construction of [1, Theorem 2.8]. It therefore carries the preceding equivalence to an equivalence of $\mathbb{E}_1$ -rings $Mg \simeq Mg_L$.

The underlying diagram of g_L is the discrete family $\{L^{\wedge j}\}_{j \geq 0}$. Consequently,

$$Mg_L \simeq \bigvee_{j \geq 0} L^{\wedge j},$$

and the multiplication induced by the operadic colimit is the usual concatenation, or tensor-algebra, multiplication. Equivalently, for every $\mathbb{E}_1$ -ring A , restriction to the weight-one summand gives a natural equivalence

$$\mathrm{Map}_{\mathrm{Alg}_{\mathbb{E}_1}(\mathrm{Sp})} \left(\bigvee_{j \geq 0} L^{\wedge j}, A \right) \simeq \mathrm{Map}_{\mathrm{Sp}}(L, A),$$

which is the universal property of $\mathbb{E}_1(L)$.

If g is an $\mathbb{E}_k$ -map, [1, Corollary 3.2] equips Mg with an $\mathbb{E}_k$ -ring structure. The construction is obtained from the same operadic colimit and is compatible with restriction along $\mathbb{E}_1 \rightarrow \mathbb{E}_k$; hence its underlying $\mathbb{E}_1$ -multiplication is the one just computed. $\square$

4.2. The odd-primary obstruction. The next proposition is the form of [6, Proposition 11.1] needed here. The proposition number refers to arXiv version 2, dated 25 June 2026. We include the proof because this is the only nonformal input in the exclusion of nonzero degrees for $n \geq 3$.

Proposition 4.3 (Rognes–Sagave–Schlichtkrull). *Let $r > 0$, and equip*

$$\mathbb{S}[x] := \mathbb{E}_1(\mathbb{S}^{2r}) \simeq \bigvee_{j \geq 0} \mathbb{S}^{2rj}$$

with its standard free $\mathbb{E}_1$ -ring structure. Then $\mathbb{S}[x]$ does not lie in the essential image of the forgetful functor $\mathrm{Alg}_{\mathbb{E}_3}(\mathrm{Sp}) \rightarrow \mathrm{Alg}_{\mathbb{E}_1}(\mathrm{Sp})$; equivalently, its standard $\mathbb{E}_1$ -ring structure admits no $\mathbb{E}_3$ -refinement.

Proof. Choose an odd prime p such that $p \nmid r$; such a prime exists because r has only finitely many prime divisors. This auxiliary prime will be used only to detect the obstruction in $\mathbb{F}_p$ -homology. Set $q = 2r$. Write

$$\mathbb{E}_k X \simeq \bigvee_{j \geq 0} C_{k,j} X$$

for the free $\mathbb{E}_k$ -ring on a spectrum X , decomposed by operadic weight. The inclusions of little-cubes operads induce maps

$$i: \mathbb{E}_1\mathbb{S}^q \longrightarrow \mathbb{E}_3\mathbb{S}^q \longrightarrow \mathbb{E}_\infty\mathbb{S}^q.$$

Suppose, for contradiction, that this $\mathbb{E}_1$ -ring admitted an $\mathbb{E}_3$ -refinement. Transporting the refinement across the given $\mathbb{E}_1$ -equivalence if necessary, we may assume that the standard $\mathbb{E}_1$ -structure on $\mathbb{E}_1\mathbb{S}^q$ is the restriction of an $\mathbb{E}_3$ -structure on the same underlying spectrum. The inclusion of the weight-one generator $\mathbb{S}^q \rightarrow \mathbb{E}_1\mathbb{S}^q$, by the universal property of the free $\mathbb{E}_3$ -ring, would extend to an $\mathbb{E}_3$ -map

$$\rho: \mathbb{E}_3\mathbb{S}^q \longrightarrow \mathbb{E}_1\mathbb{S}^q.$$

The composite ρi is an $\mathbb{E}_1$ -endomorphism of the free $\mathbb{E}_1$ -ring that is the identity on the generator. Hence $\rho i \simeq \text{id}$, so i would admit a retraction.

We work in homology with coefficients in $\mathbb{F}_p$. Let $e \in H_q(\mathbb{S}^q; \mathbb{F}_p)$ be the fundamental class. In operadic weight p , the canonical maps are

$$\mathbb{S}^{pq} \simeq C_{1,p}\mathbb{S}^q \longrightarrow C_{3,p}\mathbb{S}^q \longrightarrow C_{\infty,p}\mathbb{S}^q.$$

Let e^p denote the bottom class in $H_{pq}(C_{1,p}\mathbb{S}^q; \mathbb{F}_p)$, and use the same notation for its images in the $\mathbb{E}_3$ - and $\mathbb{E}_\infty$ -weight- p pieces. In the last of these spaces this class is the Dyer–Lashof class $Q^r(e)$. The bottom homology group in each of the three weight spaces is one-dimensional, and the displayed maps carry bottom generator to bottom generator. In particular, $H_{pq}(C_{3,p}\mathbb{S}^q; \mathbb{F}_p) \cong \mathbb{F}_p$. These facts are stated and used in the proof of [6, Proposition 11.1].

Cohen’s top operation for $\mathbb{E}_3$ -algebras gives a class

$$\xi_2(e) \in H_{pq+2p-2}(C_{3,p}\mathbb{S}^q; \mathbb{F}_p).$$

Under the map $C_{3,p}\mathbb{S}^q \rightarrow C_{\infty,p}\mathbb{S}^q$, it is carried to $Q^{r+1}(e)$. Let P_*^1 denote the homology operation dual to the first reduced power. Naturality and the odd-primary Nishida relation give

$$P_*^1 Q^{r+1}(e) = r Q^r P_*^0(e) = r e^p; \quad (4.2)$$

see [2, Theorem IV.1.1(6)] and the proof of [6, Proposition 11.1]. Because $p \nmid r$, the right-hand side is nonzero. Therefore the image of $P_*^1 \xi_2(e)$ in $H_{pq}(C_{\infty,p}\mathbb{S}^q; \mathbb{F}_p)$ is nonzero, and hence $P_*^1 \xi_2(e) \neq 0$ already in $H_{pq}(C_{3,p}\mathbb{S}^q; \mathbb{F}_p)$.

The latter group is one-dimensional, generated by $i_*(e^p)$. Thus

$$P_*^1 \xi_2(e) = \lambda i_*(e^p)$$

for some $\lambda \in \mathbb{F}_p^\times$. On the other hand, as a spectrum,

$$\mathbb{E}_1\mathbb{S}^q \simeq \bigvee_{m \geq 0} \mathbb{S}^{qm}.$$

Let $j_m: \mathbb{S}^{qm} \rightarrow \bigvee_{j \geq 0} \mathbb{S}^{qj}$ be the inclusion of the m -th summand, and let $u_m \in H_{qm}(\mathbb{S}^{qm}; \mathbb{F}_p)$ be its fundamental class. Since P_*^1 lowers degree by $2(p-1)$,

$$P_*^1(u_m) \in H_{qm-2(p-1)}(\mathbb{S}^{qm}; \mathbb{F}_p) = 0.$$

Naturality therefore gives

$$P_*^1(j_m)_*(u_m) = (j_m)_*P_*^1(u_m) = 0.$$

The classes $(j_m)_*(u_m)$, for $m \geq 0$, form a basis of $H_*(\mathbb{E}_1\mathbb{S}^q; \mathbb{F}_p)$; hence P_*^1 vanishes on the entire homology of $\mathbb{E}_1\mathbb{S}^q$. If the retraction ρ existed, naturality and $\rho_*i_* = \text{id}$ would imply

$$0 = P_*^1\rho_*\xi_2(e) = \rho_*P_*^1\xi_2(e) = \lambda\rho_*i_*(e^p) = \lambda e^p \neq 0,$$

a contradiction. This proves the proposition. $\square$

4.3. Exclusion of nonzero degree for $n \geq 3$.

Theorem 4.4. *If $n \geq 3$, every $\mathbb{E}_n$ -map $f: \mathbb{Z} \rightarrow \text{Pic}(\mathbb{S})$ has degree zero.*

Proof. By restriction along $\mathbb{E}_3 \rightarrow \mathbb{E}_n$, it is enough to treat $n = 3$. Let f be an $\mathbb{E}_3$ -map of degree d . Forgetting to $\mathbb{E}_2$ and applying proposition 3.3, write $d = 2r$. If $r < 0$, precompose with the $\mathbb{E}_\infty$ -automorphism $-1: \mathbb{Z} \rightarrow \mathbb{Z}$. Thus, if $d \neq 0$, we may assume $r > 0$.

Restrict f along the $\mathbb{E}_\infty$ -map $\mathbb{N} \hookrightarrow \mathbb{Z}$. This gives an $\mathbb{E}_3$ -map

$$g: \mathbb{N} \longrightarrow \text{Pic}(\mathbb{S}), \quad g(1) \simeq \mathbb{S}^{2r}.$$

By [1, Corollary 3.2], the Thom spectrum Mg is an $\mathbb{E}_3$ -ring. By lemma 4.2, its underlying $\mathbb{E}_1$ -ring is equivalent to

$$\mathbb{E}_1(\mathbb{S}^{2r}) \simeq \bigvee_{j \geq 0} \mathbb{S}^{2rj}$$

with its standard free $\mathbb{E}_1$ -multiplication. The essential image of the forgetful functor from $\mathbb{E}_3$ -rings to $\mathbb{E}_1$ -rings is closed under equivalence. Transporting the $\mathbb{E}_3$ -structure on Mg across the preceding equivalence would therefore extend the standard free $\mathbb{E}_1$ -structure to an $\mathbb{E}_3$ -structure. This contradicts proposition 4.3. Hence $d = 0$. $\square$

Corollary 4.5. *For every $n \geq 3$, the entire mapping space is the degree-zero fiber, and the constant map determines a canonical equivalence*

$$\mathcal{M}_n \simeq \mathcal{T}_n.$$

Proof. By theorem 4.4, every component has degree zero. The degree-zero fiber in proposition 2.6 is canonically

$$\Omega\mathcal{Y}_n \simeq \Omega^\infty F(\Sigma^\infty K(\mathbb{Z}, n), \Sigma^{n+1} \text{gl}_1(\mathbb{S})) = \mathcal{T}_n.$$

The constant map is its canonical basepoint. $\square$

5. COMPLETION OF THE FINITE- n CLASSIFICATION

Proof of theorem 1.1. The equivalences in parts (i)–(iii) are propositions 2.4 and 2.6. The degree image is determined as follows: proposition 3.1 gives $D_1 = \mathbb{Z}$, propositions 3.3 and 3.4 give $D_2 = 2\mathbb{Z}$, and theorem 4.4 gives $D_n = \{0\}$ for $n \geq 3$. The explicit mapping space decompositions are proposition 3.1 and corollaries 3.5 and 4.5. $\square$

Proof of corollary 1.2. By (1.3) and the function-spectrum adjunction,

$$\begin{aligned}\pi_j \mathcal{T}_n &\cong \pi_j F(\Sigma^\infty K(\mathbb{Z}, n), \Sigma^{n+1} \mathrm{gl}_1(\mathbb{S})) \\ &\cong [\Sigma^\infty K(\mathbb{Z}, n), \Sigma^{n+1-j} \mathrm{gl}_1(\mathbb{S})] \\ &= \widetilde{\mathrm{gl}_1(\mathbb{S})}^{n+1-j}(K(\mathbb{Z}, n)).\end{aligned}$$

This proves (1.13). The kernel of the degree homomorphism on components is G_n , because its zero fiber is $\mathcal{T}_n$. For $n \geq 3$, the degree image is zero, so $\pi_0 \mathcal{M}_n \cong G_n$. For $n = 1$, proposition 3.1 gives $\pi_0 \mathcal{M}_1 \cong \mathbb{Z}$.

For $n = 2$, the assignment

$$k \longmapsto [\Phi \circ [k]]$$

is a section of the degree map over $2\mathbb{Z}$. It is a homomorphism: the $\mathbb{E}_2$ -monoidal structure of Φ coherently identifies the pointwise smash product of $\Phi \circ [k]$ and $\Phi \circ [\ell]$ with $\Phi \circ [k + \ell]$. Hence the extension of abelian groups splits, giving $\pi_0 \mathcal{M}_2 \cong 2\mathbb{Z} \oplus G_2$. $\square$

Remark 5.1 (The residual generalized cohomology groups). The uniform classification naturally stops at the generalized cohomology of $K(\mathbb{Z}, n)$ with coefficients in $\mathrm{gl}_1(\mathbb{S})$. Choose a countable CW model and an exhaustive sequence of finite subcomplexes

$$X_0 \subset X_1 \subset \cdots \subset K(\mathbb{Z}, n), \quad \operatorname{colim}_m X_m = K(\mathbb{Z}, n).$$

For each m , the Atiyah–Hirzebruch spectral sequence has the form

$$E_2^{s,t}(m) = \tilde{H}^s(X_m; \pi_t \mathrm{gl}_1(\mathbb{S})) \implies \widetilde{\mathrm{gl}_1(\mathbb{S})}^{s-t}(X_m),$$

where

$$\pi_t \mathrm{gl}_1(\mathbb{S}) = \begin{cases} \mathbb{Z}/2, & t = 0, \\ \pi_t \mathbb{S}, & t > 0. \end{cases}$$

Indeed,

$$\Sigma^\infty K(\mathbb{Z}, n) \simeq \operatorname{colim}_m \Sigma^\infty X_m,$$

so mapping into a spectrum converts this filtered colimit into an inverse limit of mapping spectra. The Milnor exact sequence for the homotopy groups of that inverse limit is

$$\begin{aligned}0 \longrightarrow \lim_m^1 \widetilde{\mathrm{gl}_1(\mathbb{S})}^{q-1}(X_m) &\longrightarrow \widetilde{\mathrm{gl}_1(\mathbb{S})}^q(K(\mathbb{Z}, n)) \\ &\longrightarrow \lim_m \widetilde{\mathrm{gl}_1(\mathbb{S})}^q(X_m) \longrightarrow 0.\end{aligned}$$

Thus inverse-limit and phantom-map phenomena must be retained. The function-spectrum formula in theorem 1.1 is exact and includes these phenomena automatically; a naive passage from the finite calculations would not.

6. THE $\mathbb{E}_\infty$ -CASE

Lemma 6.1. *The zeroth space of the endomorphism spectrum of $H\mathbb{Z}$ is discrete:*

$$\Omega^\infty F(H\mathbb{Z}, H\mathbb{Z}) \simeq \mathbb{Z}_{\text{disc}}.$$

Proof. The standard t -structure on spectra has heart equivalent to the category of abelian groups, under $A \mapsto HA$. Therefore

$$\pi_0 F(H\mathbb{Z}, H\mathbb{Z}) \cong \text{Hom}_{\mathbf{Ab}}(\mathbb{Z}, \mathbb{Z}) \cong \mathbb{Z}.$$

For $i > 0$,

$$\pi_i F(H\mathbb{Z}, H\mathbb{Z}) = [\Sigma^i H\mathbb{Z}, H\mathbb{Z}] = 0$$

by t -structure orthogonality, since $\Sigma^i H\mathbb{Z} \in \text{Sp}_{\geq 1}$ and $H\mathbb{Z} \in \text{Sp}_{\leq 0}$. Thus the zeroth space is zero-truncated with components indexed by $\mathbb{Z}$. $\square$

Proof of theorem 1.3. Infinite-loop-space recognition identifies grouplike $\mathbb{E}_\infty$ -spaces with connective spectra; see [3, Section 1.4, p. 132] and [3, Remark 5.2.6.26]. Under this equivalence, $\mathbb{Z}$ corresponds to $H\mathbb{Z}$, while $\text{Pic}(\mathbb{S})$ corresponds to $\text{pic}(\mathbb{S})$. Since connective spectra form a full subcategory of spectra, its derived mapping space is

$$\mathcal{M}_\infty \simeq \Omega^\infty F(H\mathbb{Z}, \text{pic}(\mathbb{S})),$$

proving (1.14).

Applying $F(H\mathbb{Z}, -)$ to (1.1) gives a fiber sequence of mapping spectra

$$F(H\mathbb{Z}, \Sigma \text{gl}_1(\mathbb{S})) \longrightarrow F(H\mathbb{Z}, \text{pic}(\mathbb{S})) \longrightarrow F(H\mathbb{Z}, H\mathbb{Z}).$$

By lemma 6.1, the zeroth space of the final term is $\mathbb{Z}_{\text{disc}}$, and the induced map on zeroth spaces is the degree map. Every $\mathbb{E}_\infty$ -map restricts to an $\mathbb{E}_3$ -map, so theorem 4.4 implies that its degree is zero. Hence the entire zeroth space of the middle term lies over the zero component of the final term. The fiber over that component is

$$\Omega^\infty F(H\mathbb{Z}, \Sigma \text{gl}_1(\mathbb{S})),$$

which proves (1.15). $\square$

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